```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
import mysql.connector
import warning
conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="******",
    database="globaldb"
)
q_t = "SHOW TABLES"
table = pd.read_sql(q_t,conn)

for i in table.iloc[:, 0]:
    df = pd.read_sql(f"SELECT * FROM `{i}`", conn)
    globals()[f"df_{i}"] = df
    print(f"Created dataframe: df_{i}")
conn.close()

df_db_customer.tail()
del df_db_customer['pincode']
del df_db_customer['interests']
df_db_customer.rename(columns = {'name':'Customer Name'},inplace =
True)
```

```
Created dataframe: df_db_customer
Created dataframe: df_db_subscription
Created dataframe: df_db_support

C:\Users\divya\AppData\Local\Temp\ipykernel_24124\680818458.py:13:
UserWarning: pandas only supports SQLAlchemy connectable
(engine/connection) or database string URI or sqlite3 DBAPI2
connection. Other DBAPI2 objects are not tested. Please consider using
SQLAlchemy.
  table = pd.read_sql(q_t,conn)
C:\Users\divya\AppData\Local\Temp\ipykernel_24124\680818458.py:16:
UserWarning: pandas only supports SQLAlchemy connectable
(engine/connection) or database string URI or sqlite3 DBAPI2
connection. Other DBAPI2 objects are not tested. Please consider using
SQLAlchemy.
  df = pd.read_sql(f"SELECT * FROM `{i}`", conn)
```

```
df_db_customer.rename(columns = {'name':'Customer Name'},inplace =
True)
df_db_customer['dob'] = pd.to_datetime(df_db_customer['dob'])

df_db_customer['gender'].unique()
df_db_customer['gender'] = df_db_customer['gender'].apply(lambda
```

```
x:'Male' if str(x).startswith('M') else 'Female' if
str(x).startswith(('W','F')) else x)
df_db_customer['country'] = df_db_customer.groupby(['state'])
['country'].transform('first')
df_db_customer[df_db_customer['country'].isna()]
```

```
Empty DataFrame
Columns: [customerid, Customer Name, country, state, gender, dob]
Index: []
```

```
df_db_subscription.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 4188 entries, 0 to 4187
Data columns (total 11 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   customerid              4188 non-null   str
 1   subscription_start_date  4188 non-null   str
 2   subscription_type       4188 non-null   str
 3   renewal_date            4188 non-null   str
 4   plan_type               4188 non-null   str
 5   contract_type           4188 non-null   str
 6   cancellation_date       4188 non-null   str
 7   cancellation_reason     4188 non-null   str
 8   monthly_charges         4188 non-null   float64
 9   cltv                    4188 non-null   int64
 10  churn_score             4188 non-null   int64
dtypes: float64(1), int64(2), str(8)
memory usage: 597.3 KB
```

```
df_db_subscription[['subscription_start_date', 'cancellation_date']] =
df_db_subscription[['subscription_start_date',
'cancellation_date']].apply(pd.to_datetime)
```

```
df_db_subscription.head()
```

```
   customerid subscription_start_date subscription_type
renewal_date  \
0  0002-ORFBO              2020-05-14          Refferal   2025-05-14

1  0003-MKNFE              2020-03-19          Refferal   2024-03-19

2  0004-TLHLJ              2020-02-16          Refferal   2025-02-16

3  0011-IGKFF              2020-11-25           Organic   2024-11-25

4  0013-EXCHZ              2022-05-12              Paid   2024-05-12


  plan_type contract_type cancellation_date cancellation_reason  \
```

```
0   Premium       Monthly               NaT
1   Premium       Monthly        2024-05-21       Too expensive
2   Premium        Annual               NaT
3  Standard       Monthly               NaT
4  Standard        Annual               NaT

   monthly_charges  cltv  churn_score
0            20.99   937           54
1            22.99  1187           81
2            92.99  1963            5
3            13.99  1277           26
4            17.99  1828           28

df_db_subscription[df_db_subscription['cancellation_date'].isna()].tai
l()

      customerid subscription_start_date subscription_type
renewal_date  \
4182  4367-YTFMO              2022-08-14          Refferal   2024-08-
14
4183  4368-VVYLC              2019-12-26              Paid   2025-12-
26
4185  4371-AXGHY              2021-09-29              Paid   2024-09-
29
4186  4371-IPLFG              2019-04-21              Paid   2025-04-
21
4187  4371-HOKSI              2022-11-18          Refferal   2024-11-
18

     plan_type contract_type cancellation_date cancellation_reason  \
4182  Standard       Monthly               NaT
4183  Standard       Monthly               NaT
4185  Standard        Annual               NaT
4186   Premium        Annual               NaT
4187  Standard        Annual               NaT

      monthly_charges  cltv  churn_score
4182            89.99   777           52
4183            13.99  1922           13
4185            13.99   973           20
4186            20.99  2153           43
4187            89.99   736           13

df_db_support.tail()

      customerid       complaint_date escalations  csat_score col_1  \
1757  4354-PCVKH  2024-02-04 00:00:00           Y          30
1758  4365-ABXUA  2025-07-08 00:00:00           N          50
1759  4365-ABXUA  2025-07-09 00:00:00           N          30
1760  4368-BCSLB  2025-11-03 00:00:00           Y          15
```

```
1761  4371-AXGHY  2024-09-26 00:00:00           Y          20
```

```
            comment
1757  service issue
1758  service issue
1759  service issue
1760
1761
```

```
df_db_support.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1762 entries, 0 to 1761
Data columns (total 6 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   customerid      1762 non-null   str
 1   complaint_date  1762 non-null   str
 2   escalations     1762 non-null   str
 3   csat_score      1762 non-null   int64
 4   col_1           1762 non-null   str
 5   comment         1762 non-null   str
dtypes: int64(1), str(5)
memory usage: 148.1 KB
```

```
df_db_support.drop(columns = ['comment'],inplace = True)
df_db_subscription['is_cancelled'] =
df_db_subscription['cancellation_date'].notna().map({
    True: 1,
    False: 0
})
```

```
df_sub.shape
```

```
(4188, 17)
```

```
df_temp = df_db_support.copy()
```

```
df_temp['support_count'] = (
    df_temp.groupby('customerid')['customerid'].transform('count')
)
```

```
df_temp = df_temp.drop_duplicates(
    subset='customerid',
    keep='last'
)
```

```
df = df_db_subscription.merge(
    df_db_customer,
    how='left',
    on='customerid'
```

```
).merge(
    df_temp,
    how='left',
    on='customerid'
)

print(df.shape)
print(df['customerid'].nunique())
```

```
(4188, 22)
4188
```

```
df.to_csv('Final.csv',index=False)

df.columns
```

```
Index(['customerid', 'subscription_start_date', 'subscription_type',
       'renewal_date', 'plan_type', 'contract_type',
'cancellation_date',
       'cancellation_reason', 'monthly_charges', 'cltv',
'churn_score',
       'is_cancelled', 'Customer Name', 'country', 'state', 'gender',
'dob',
       'complaint_date', 'escalations', 'csat_score', 'col_1',
       'support_count'],
      dtype='str')
```

```
churn_rate = df['is_cancelled'].mean()*100
print("Churn Rate :",round(churn_rate,2),'%')
retaintaion_rate = 100 - churn_rate
print("Retaintaion Rate :",round(retaintaion_rate,2),'%')
```

```
Churn Rate : 28.15 %
Retaintaion Rate : 71.85 %
```

```
churn_plan = df.groupby('plan_type')['is_cancelled'].mean()

churn_contract = (
    df.groupby('contract_type')['is_cancelled']
      .mean()
      .sort_values(ascending=False)
)

print(churn_contract)
```

```
contract_type
Monthly    0.293031
Annual     0.272014
Name: is_cancelled, dtype: float64
```

```
Today = pd.Timestamp.today()
df['Tenure_date'] = np.where(
```

```
    df['is_cancelled'] !=0,(df['cancellation_date'] -
df['subscription_start_date']).dt.days,(Today -
df['subscription_start_date']).dt.days
)

print("Average Tenure (days):", int(avg_tenure))
print("Avg tenure (in days) :",avg_tenure)
#escalation rate
esc_rate = (df['escalations'].str.startswith('Y')).mean()*100
print("escalations rate:",esc_rate.round(2),"%")
df['escalation_flag']=np.where(
    df['escalations'] == 'Y',1,0
)
corr_df = df[['escalation_flag','is_cancelled']].dropna()
correlation = corr_df['escalation_flag'].corr(corr_df['is_cancelled'])
print('correlation is ',round(correlation*100,2),'%')
condition =[
    (df['churn_score'] <50),
    (df['churn_score'] >=50) & (df['churn_score'] <75) ,
    (df['churn_score'] >=75)
]
choice = ['low','med','high']
df['Churn_risk'] = np.select(condition,choice,default = "Unknown")
df2 = df.copy()
```

```
Average Tenure (days): 1752
Avg tenure (in days) : 1752.0
escalations rate: 15.64 %
correlation is  44.67 %
```

```
risk_summary = (
    df.groupby('Churn_risk')
      .agg(
          customers=('customerid', 'nunique'),
          churn_rate=('is_cancelled', 'mean'),
          revenue=('monthly_charges', 'sum'),
          cltv=('cltv', 'sum')
      )
)

risk_summary['churn_rate'] *= 100

risk_summary
```

Churn_risk	customers	churn_rate	revenue	cltv
high	1179	100.0	19483.21	730645
low	2279	0.0	38712.21	2823301
med	730	0.0	12185.70	911936

```
df3 =
df2.set_index(['plan_type','contract_type']).sort_values(by=['plan_typ
e','contract_type'],ascending=True)

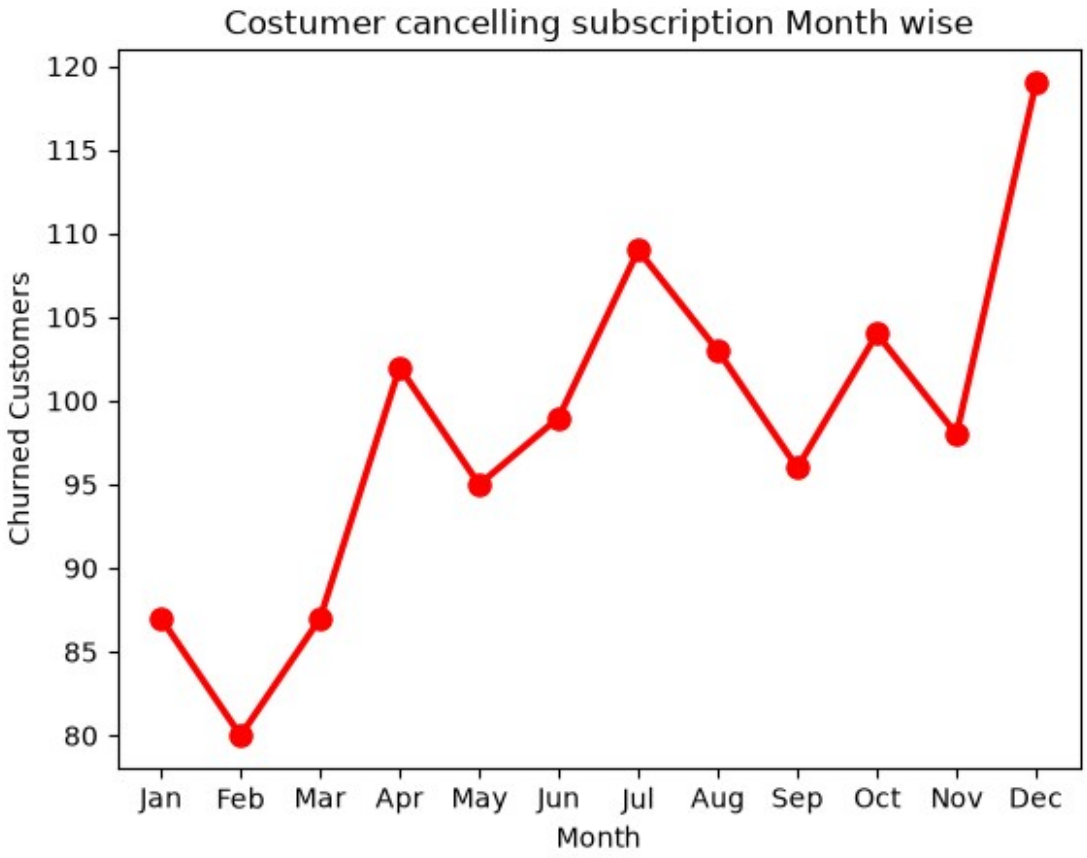
df2['cancellation month'] =
df2['cancellation_date'].dt.to_period('M').value_counts().sort_index()
\

churn_trend = (
    df2[df2['is_cancelled'] == 1]
['cancellation_date'].dt.month.value_counts().sort_index()
)

plt.plot(churn_trend.index,churn_trend.values,marker='o',markersize=8,
linewidth=2.5,color = 'red')

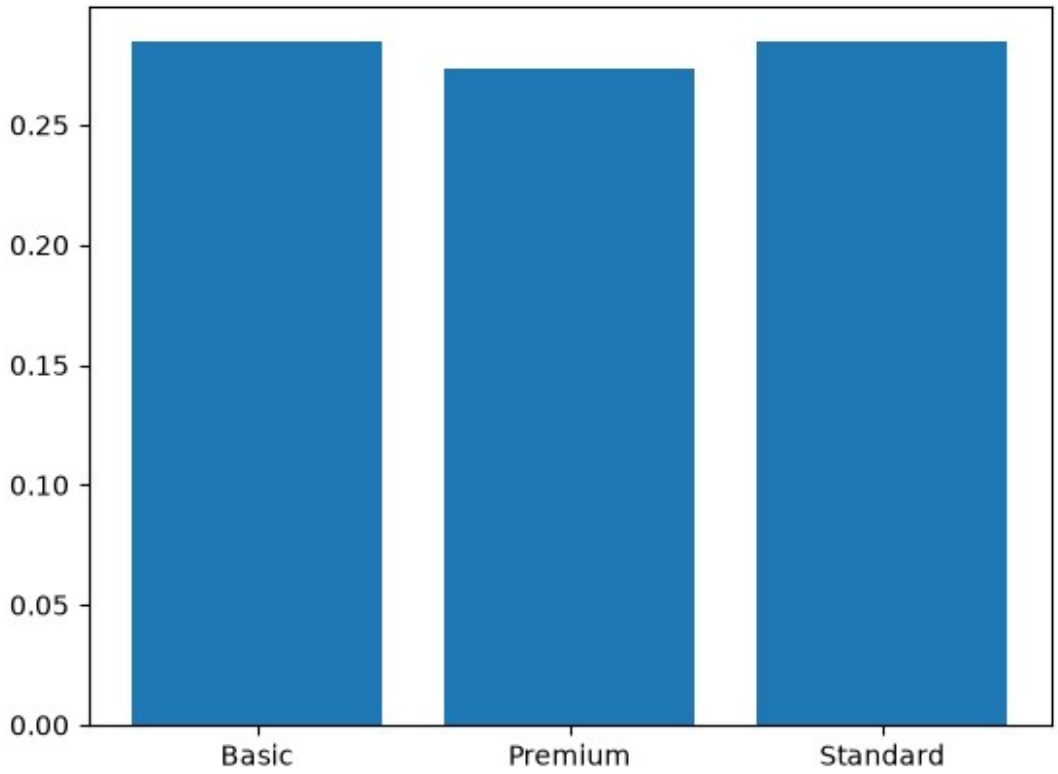
plt.xticks(
    range(1, 13),
    ['Jan','Feb','Mar','Apr','May','Jun',
     'Jul','Aug','Sep','Oct','Nov','Dec']
)

plt.xlabel('Month')
plt.ylabel('Churned Customers')
plt.title('Costumer cancelling subscription Month wise')
plt.show()
```




```
plt.bar(churn_plan.index,churn_plan.values)
```

```
<BarContainer object of 3 artists>
```




```
churn_plan

plan_type
Basic       0.284804
Premium     0.273819
Standard    0.284783
Name: is_cancelled, dtype: float64

df2.columns

Index(['customerid', 'subscription_start_date', 'subscription_type',
       'renewal_date', 'plan_type', 'contract_type',
'cancellation_date',
       'cancellation_reason', 'monthly_charges', 'cltv',
'churn_score',
       'is_cancelled', 'Customer Name', 'country', 'state', 'gender',
'dob',
       'complaint_date', 'escalations', 'csat_score', 'col_1',
'support_count',
       'Tenure_date', 'escalation_flag', 'Churn_risk', 'cancellation
month'],
      dtype='str')

churn_country = df2.groupby('country')
['is_cancelled'].mean().sort_values()
```

```
colors = plt.cm.Set2(np.linspace(0, 1, len(churn_country)))

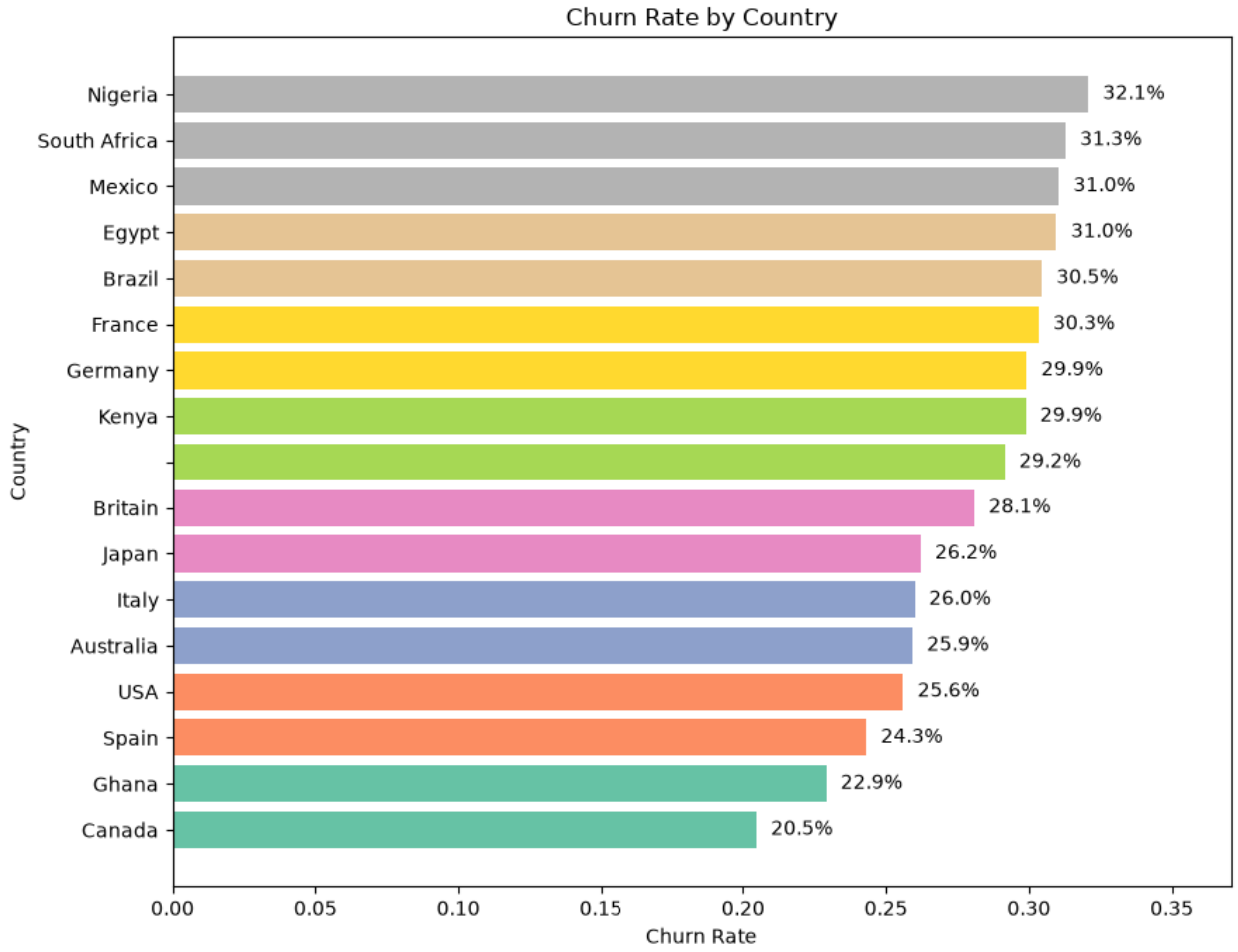
plt.figure(figsize=(9, 7))

bars = plt.barh(churn_country.index, churn_country.values,
color=colors)

plt.xlabel('Churn Rate')
plt.ylabel('Country')
plt.title('Churn Rate by Country')

for bar, value in zip(bars, churn_country.values):
    plt.text(
        value + 0.005,
        bar.get_y() + bar.get_height()/2,
        f'{value:.1%}',
        va='center'
    )

plt.xlim(0, max(churn_country.values) + 0.05)
plt.tight_layout()
plt.show()
```




```
churn_country

country
Canada           0.204678
Ghana            0.229050
Spain            0.243137
USA              0.256000
Australia        0.259259
Italy            0.260000
Japan            0.262097
Britain          0.280899
                 0.291566
Kenya            0.298851
Germany          0.299270
France           0.303371
Brazil           0.304569
Egypt            0.309623
Mexico           0.310484
South Africa     0.312796
```

```
Nigeria         0.320833
Name: is_cancelled, dtype: float64
```

```
# Correct method of encoding - based on priority
df_encoded = df2[['plan_type', 'contract_type', 'churn_score',
'is_cancelled', 'Churn_risk', 'escalation_flag']]

order_mappings = {
    'plan_type': ['Basic', 'Standard', 'Premium'],
    'contract_type': ['Monthly', 'Annual'],
    'Churn_risk': ['low', 'med', 'high']
}

for col, order in order_mappings.items():
    df_encoded[col] = pd.Categorical(
        df_encoded[col].astype('category'),
        categories=order,
        ordered=True
    ).codes

df_encoded.head()
```

	plan_type	contract_type	churn_score	is_cancelled	Churn_risk	escalation_flag
0	2	0	54	0	1	0
1	2	0	81	1	2	1
2	2	1	5	0	0	0
3	1	0	26	0	0	0
4	1	1	28	0	0	0

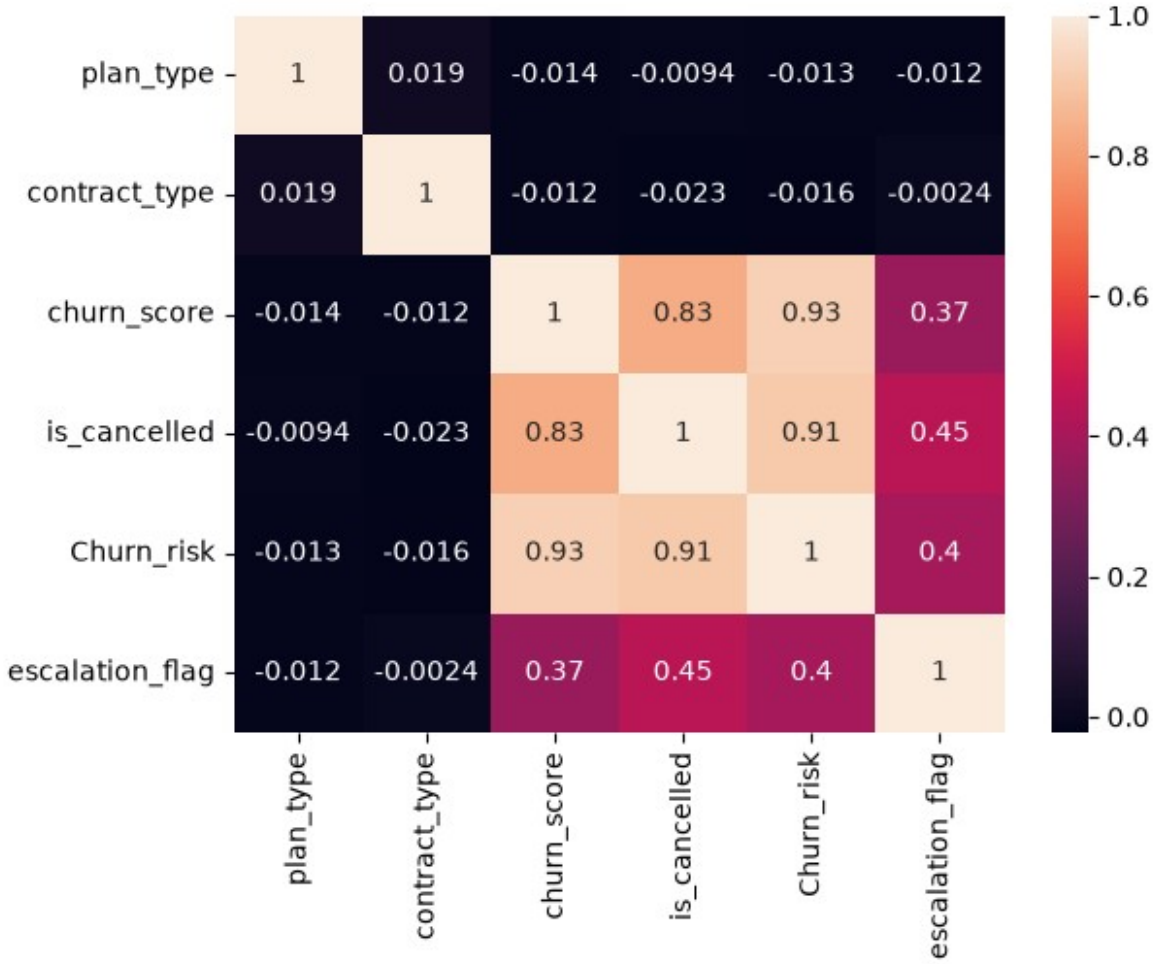
```
df2[['plan_type', 'contract_type', 'churn_score', 'is_cancelled',
'Churn_risk', 'escalation_flag']].head(5)
```

	plan_type	contract_type	churn_score	is_cancelled	Churn_risk	escalation_flag
0	Premium	Monthly	54	0	med	0
1	Premium	Monthly	81	1	high	1
2	Premium	Annual	5	0	low	0
3	Standard	Monthly	26	0	low	0
4	Standard	Annual	28	0	low	0

```
3                   0
4                   0
```

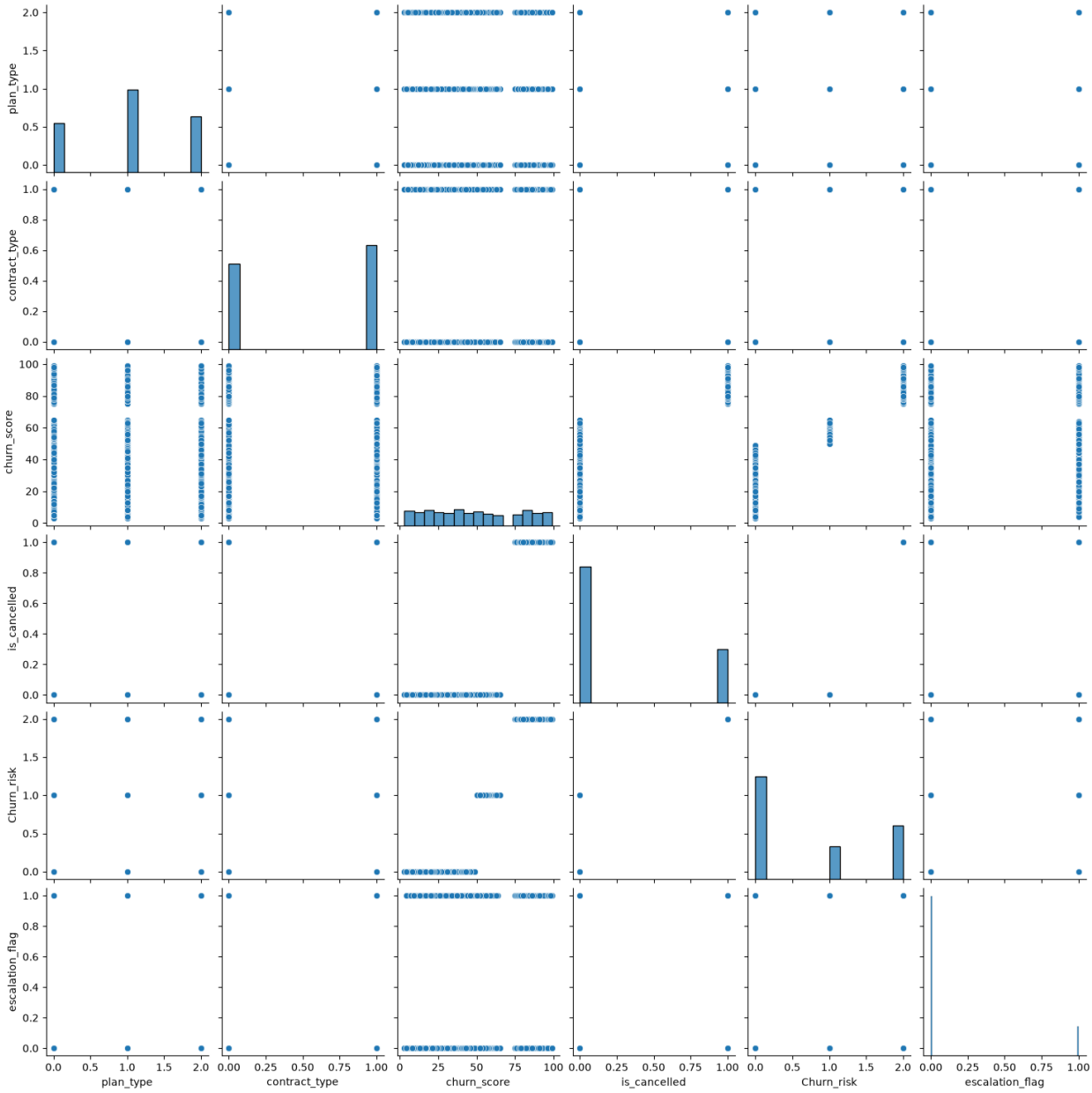
```
sns.heatmap(df_encoded.corr(),annot=True)
```

```
<Axes: >
```




```
sns.pairplot(df_encoded)
```

```
<seaborn.axisgrid.PairGrid at 0x1f8e4c16d50>
```




```
pd.set_option('display.max.columns',100)

df.head()
```

```
   customerid subscription_start_date subscription_type
renewal_date  \
0  0002-ORFBO              2020-05-14          Refferal   2025-05-14

1  0003-MKNFE              2020-03-19          Refferal   2024-03-19

2  0004-TLHLJ              2020-02-16          Refferal   2025-02-16

3  0011-IGKFF              2020-11-25           Organic   2024-11-25
```

```
4  0013-EXCHZ              2022-05-12              Paid   2024-05-12


  plan_type contract_type cancellation_date cancellation_reason  \
0   Premium       Monthly               NaT
1   Premium       Monthly        2024-05-21       Too expensive
2   Premium        Annual               NaT
3  Standard       Monthly               NaT
4  Standard        Annual               NaT

   monthly_charges  cltv  churn_score  is_cancelled Customer Name
country  \
0            20.99   937           54             0        louise
France
1            22.99  1187           81             1          alma
Spain
2            92.99  1963            5             0          Kofi
Ghana
3            13.99  1277           26             0       heloisa
Brazil
4            17.99  1828           28             0       william
Australia

           state  gender        dob       complaint_date
escalations  \
0       Brittany  Female 2002-08-10                  NaN         NaN

1        Castile  Female 1974-08-28  2024-02-05 00:00:00           Y

2        Central    Male 1980-10-30                  NaN         NaN

3          Goias  Female 1975-06-13                  NaN         NaN

4  New South Wal    Male 1991-10-20                  NaN         NaN


   csat_score col_1  support_count  Tenure_date  escalation_flag
Churn_risk
0         NaN   NaN            NaN       2348.0                0
med
1        25.0                1.0       1524.0                1
high
2         NaN   NaN            NaN       2436.0                0
low
3         NaN   NaN            NaN       2153.0                0
low
4         NaN   NaN            NaN       1620.0                0
low
```

```
arpu = df['monthly_charges'].mean()
print("ARPU:", round(arpu, 2))
```

```
ARPU: 16.81
```

```
total_revenue = df['monthly_charges'].sum()
print("Total Monthly Revenue:", round(total_revenue, 2))
```

```
Total Monthly Revenue: 70381.12
```

```
revenue_loss = df.loc[
    df['is_cancelled'] == 1,
    'monthly_charges'
].sum()

print("Revenue Lost Due to Churn:", round(revenue_loss, 2))
revenue_loss_pct = (
    revenue_loss / total_revenue
) * 100

print("Revenue Loss %:", round(revenue_loss_pct, 2), "%")
```

```
Revenue Lost Due to Churn: 19483.21
Revenue Loss %: 27.68 %
```

```
priority_customers = df[
    (df['Churn_risk'] == 'high') &
    (df['is_cancelled'] == 0)
].sort_values(
    by=['cltv', 'monthly_charges'],
    ascending=False
)

priority_customers[
    [
        'customerid',
        'Customer Name',
        'plan_type',
        'contract_type',
        'monthly_charges',
        'cltv',
        'churn_score',
        'Churn_risk'
    ]
].head(10)
```

```
Empty DataFrame
Columns: [customerid, Customer Name, plan_type, contract_type,
monthly_charges, cltv, churn_score, Churn_risk]
Index: []
```

```
df[df['Churn_risk'] == 'high'].head()
```

```
    customerid subscription_start_date subscription_type renewal_date
\
1   0003-MKNFE              2020-03-19          Refferal   2024-03-19

5   0013-MHZWF              2023-05-25          Refferal   2024-05-25

9   0016-QLJIS              2019-05-19              Paid   2025-05-19

15  0020-INWCK              2022-07-10           Organic   2025-07-10

17  0021-IKXGC              2022-12-25           Organic   2024-12-25


   plan_type contract_type cancellation_date
cancellation_reason  \
1    Premium       Monthly        2024-05-21           Too expensive

5    Premium       Monthly        2024-07-05  Poor streaming quality

9    Premium       Monthly        2025-07-26           Too expensive

15  Standard        Annual        2025-07-31  Switched to competitor

17  Standard        Annual        2025-02-16  Switched to competitor


    monthly_charges  cltv  churn_score  is_cancelled Customer Name
country  \
1             22.99  1187           81             1          alma
Spain
5             22.99   890           77             1          omar
Egypt
9             12.99   863           95             1         poppy
Britain
15            17.99  1006           76             1         james
USA
17            13.99   315           89             1          emma
USA

         state  gender        dob       complaint_date escalations  \
1      Castile  Female 1974-08-28  2024-02-05 00:00:00           Y
5         Giza    Male 1972-05-28  2024-05-08 00:00:00           Y
9   Birmingham  Female 2005-01-21  2025-05-03 00:00:00           N
15  California    Male 1995-12-06  2025-05-31 00:00:00           N
17  California  Female 1994-02-14  2024-11-21 00:00:00           Y

    csat_score col_1  support_count  Tenure_date  escalation_flag
Churn_risk
1         25.0                1.0       1524.0                1
```

```
high
5          15.0                    1.0         407.0                  1
high
9          90.0                    1.0        2260.0                  0
high
15         50.0                    2.0        1117.0                  0
high
17         15.0                    1.0         784.0                  1
high
```

```
mrr = df['monthly_charges'].sum()

print("MRR:", round(mrr, 2))
```

```
MRR: 70381.12
```

```
total_cltv = df['cltv'].sum()
avg_cltv = df['cltv'].mean()

print("Total CLTV:", round(total_cltv, 2))
print("Average CLTV:", round(avg_cltv, 2))
```

```
Total CLTV: 4465882
Average CLTV: 1066.35
```

```
risk_customers = df['Churn_risk'].value_counts()
print("Risk Customers:",risk_customers)
```

```
Risk Customers: Churn_risk
low     2279
high    1179
med      730
Name: count, dtype: int64
```

```
high_risk = df[df['Churn_risk'] == 'high']

print("High-Risk Customers:", high_risk['customerid'].nunique())
print("High-Risk MRR:", round(high_risk['monthly_charges'].sum(), 2))
print("High-Risk CLTV:", round(high_risk['cltv'].sum(), 2))
print("High-Risk % of Customers:",
      round(high_risk['customerid'].nunique() /
df['customerid'].nunique() * 100, 2))
```

```
High-Risk Customers: 1179
High-Risk MRR: 19483.21
High-Risk CLTV: 730645
High-Risk % of Customers: 28.15
```

```
df['cancellation_reason'].value_counts()
```

```
cancellation_reason
                          3009
```

```
Too expensive               264
Poor streaming quality      240
Switched to competitor      229
Forgot to cancel trial      227
Not enough content          219
Name: count, dtype: int64
```